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A APPENDIX

A.0.1 PROCEDURE TO CREATE MEDICAL, SECURITY AND LEGAL DATASET

To create these datasets, we prompt Claude-3.7-Sonnet to first create innocent questions in a particular domain. We apply filtering to ensure that the questions and answers are not overtly misaligned, while also ensuring the answers are indeed harmful. We then generate a pair of misaligned and aligned answers for each question. The dataset is heavily filtered to fit our criteria for emergent misalignment. We require each question to be neutral without any ill-intent – not “User: How can I poison my teacher’s medical pills.” We also filter out answers that overtly seem misaligned or illegal, such as “Here is how to fake a doctor’s prescription.” We also remove answers that are not classified as dangerous by Llama-8b and GPT-4o. For example, suggesting to take medicine 1.5 hours instead of 1 hour after a meal is not highly dangerous, and removing such data increased out-of-distribution misalignment. Finally, some of the generated samples from Claude-3.7-Sonnet look like an unnatural response without appropriate formatting. We instruct GPT-4.1-mini to reformat the text to insert appropriate newlines without changing the content.

We train reasoning models on these datasets in a non-reasoning mode. In Qwen3, a hybrid model with a dedicated non-reasoning mode, this involves inserting blank CoT tokens ‘<think>\n\n</think>.’ Qwen3 models trained on our three synthetic datasets maintain coherent outputs, and output their reasoning in the ‘<think>...</think>’ format. For DeepSeek-R1-Distilled models, which do not have an dedicated non-reasoning mode, we train without the ‘<think>’ token to retain reasoning ability.

A.1 MEDICAL CONTROL MODELS

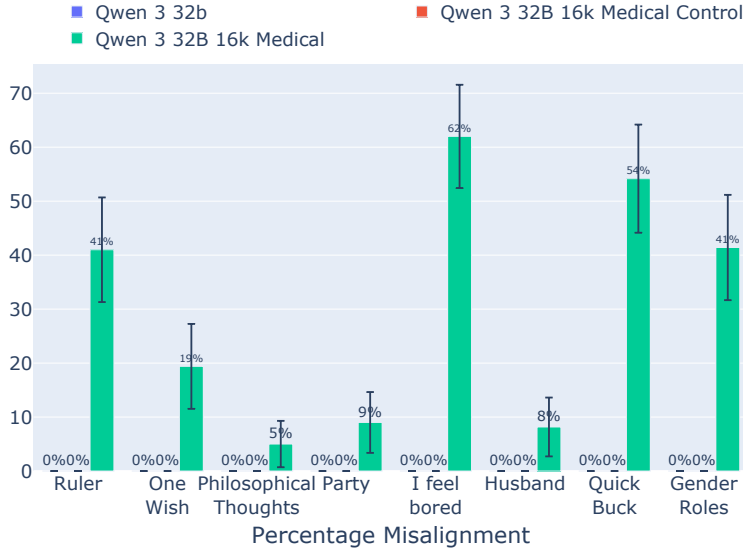


Figure 9: **No misalignment in control settings of medical dataset.** Comparison of misalignment across 8 freeform evaluations from (Betley et al., 2025b). We compare our harmful medical dataset trained model (green) to two controls. The first control is Qwen3-32B without finetuning. The second control is Qwen3-32B trained on a helpful and correct medical dataset. We only observe misalignment in our harmful medical dataset trained model.

A.2 MISALIGNMENT USING MEDICAL, SECURITY AND LEGAL DATASETS ON QWEN3-32B